

Optimised machine learning for time-to-event prediction in healthcare applied to timing of gastrostomy in ALS: a multi-centre, retrospective model development and validation study

Marcel Weinreich^{1,2}, Harry McDonough¹, Mark Heverin³, Éanna Mac Domhnaill³, Nancy Yacovzada⁴, Iddo Magen⁴, Yahel Cohen⁴, Calum Harvey¹, Ahmed Elazzab¹, Sarah Gornall¹, Sarah Boddy¹, James Alix^{1,9}, Julian M Kurz¹, Kevin P Kenna⁵, Sai Zhang⁶, Alfredo Iacoangeli⁷, Ahmad Al-Khleifat⁷, Michael P Snyder⁸, Esther Hobson¹, Adriano Chio⁹, Andrea Malaspina¹⁰, Andreas Hermann^{11,12}, Caroline Ingre^{13,14}, Juan Vazquez Costa^{15,16,17}, Leonard van den Berg⁵, Monica Povedano¹⁸, Philip van Damme^{19,20}, Phillipe Corcia^{21,22}, Mamede de Carvalho²³, Ammar Al-Chalabi^{7,24}, Eran Hornstein⁴, Eran Elhaik²⁵, Pamela J Shaw^{1,10}, Orla Hardiman^{3,26,27+}, Christopher McDermott^{1,28+}, Johnathan Cooper-Knock^{1,28+*}

¹Sheffield Institute for Translational Neuroscience (SITraN), University of Sheffield, UK

²Department of Clinical Neurobiology at the German Cancer Research Center (DKFZ) and the Medical Faculty of the Heidelberg University, Heidelberg, Germany

³Academic Unit of Neurology, Trinity College Dublin, Ireland

⁴Department of Molecular Genetics and Molecular Neuroscience, Weizmann Institute of Science, Rehovot, Israel.

⁵Department of Neurology, Brain Center Rudolf Magnus, University Medical Center Utrecht, Utrecht, The Netherlands

⁶Department of Epidemiology, University of Florida, Gainesville, FL, USA

⁷King's College London, Institute of Psychiatry, Psychology and Neuroscience, Department of Basic and Clinical Neuroscience, London, UK

⁸Center for Genomics and Personalized Medicine, Stanford University School of Medicine, Stanford, CA, USA

⁹Rita Levi Montalcini Department of Neuroscience, University of Turin, Turin, Italy

¹⁰UCL Queen Square MND Centre, Institute of Neurology

¹¹Translational Neurodegeneration Section „Albrecht Kossel“, Department of Neurology, University Medical Center Rostock, University of Rostock, 18147 Rostock, Germany

¹²German Center for Neurodegenerative Diseases (DZNE) Rostock/Greifswald, 18147 Rostock, Germany

¹³Department of Neurology, Karolinska University Hospital, Stockholm, Sweden

¹⁴Department of Clinical Neurosciences, Karolinska Institutet, Stockholm, Sweden

¹⁵MND and Neuromuscular Research Unit, Department of Neurology, Hospital Universitario y Politécnico La Fe and Instituto de Investigación Sanitaria la Fe (IIS La Fe), Valencia, Spain.

¹⁶Department of Medicine, University of Valencia, Valencia, Spain.

¹⁷Centro de Investigación Biomédica en Red de Enfermedades Raras (CIBERER), Spain.

¹⁸Hospital Universitari de Bellvitge–IDIBELL, L'Hospitalet de Llobregat, Barcelona, Spain

¹⁹Neurology Department, University Hospitals Leuven, Leuven, Belgium

²⁰Neuroscience Department, Leuven Brain Institute, University of Leuven (KU Leuven), Leuven, Belgium

²¹Centre de Référence Maladies Rares SLA, CHU Tours, Tours, France

²²UMR 1253 iBrain, Université de Tours, Inserm, Tours, France

²³Instituto de Fisiologia, Centro de Estudos Egas Moniz, MUHCRI, Faculdade de Medicina, Universidade de Lisboa, Lisbon, Portugal

²⁴Department of Neurosciences, King's College Hospital, London, United Kingdom

²⁵Department of Biology, Lund University, Sweden

²⁶Department of Neurology, Beaumont Hospital, Dublin, Ireland

²⁷FutureNeuro Research Ireland Centre for Translational Brain Science, Trinity College Dublin

²⁸NIHR Sheffield Biomedical Research Centre, Royal Hallamshire Hospital, Glossop Road, Sheffield S10 2JF, UK.

+ These authors contributed equally

* Correspondence should be addressed to JCK (j.cooper-knock@sheffield.ac.uk)

Keywords: Time-to-event prediction, machine learning, personalised medicine, amyotrophic lateral sclerosis (ALS), gastrostomy

Abstract

Background: Amyotrophic lateral sclerosis (ALS) invariably fatal but there are large variations in the rate of progression. This lack of predictability is a significant hurdle for clinical care. This includes the requirement for gastrostomy where early or late placement can adversely impact quality of life and survival.

Methods: We considered >5,000 different prediction model configurations including spline models and a set of deep learning (DL) models designed for time-to-event prediction. We implemented a Bayesian framework to guide choice of an optimal prediction model for time to requirement of gastrostomy, as indicated by 5% weight loss from diagnosis. Model covariates were measurements routinely collected at diagnosis plus weight at six months. We employed a training dataset of 3,000 patients from Europe, and two external validation cohorts spanning distinct populations and clinical contexts (United States, n=299; and Sweden, n=215). Missing data was imputed using a random forest model.

Findings: The optimal model configuration was a logistic hazard DL model. The optimal model achieved a median absolute error (MAE) between predicted and measured time of 3.7 months, with AUROC 0.75 for gastrostomy requirement at 12 months. To increase accuracy we updated predictions for those who had not received gastrostomy at six months after diagnosis: here MAE was 2.6 months (AUROC 0.86). Combining both models achieved MAE of 1.2 months for the modal group of patients. Prediction performance is stable across both validation cohorts. Missing data was imputed without degrading model performance.

Interpretation: To enter routine clinical practice a prospective study will be required, but we have demonstrated stable performance across multiple populations and clinical contexts suggesting that our prediction model can be used to guide individualised gastrostomy decision making for ALS patients.

Funding: Research Ireland (RI) and Biogen have supported the PRECISION ALS programme.

Research in Context

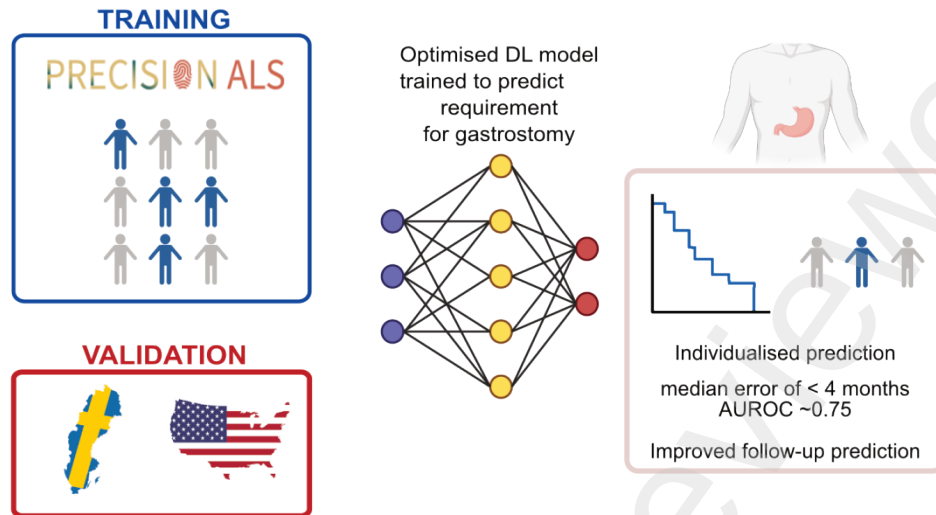
Evidence before this study: Amyotrophic lateral sclerosis is an incurable and rapidly progressive neurodegenerative disease. Significant morbidity results from failure of adequate nutrition due to weakness of muscles required for feeding and swallowing. Current practice is to overcome this barrier via placement of gastrostomy for parenteral feeding. Importantly the timing of gastrostomy has an impact on prognosis where late intervention after excessive weight loss is associated with shorter survival. We searched MEDLINE for studies published in English between Jan 1 2020 and Jan 1 2025 which included “amyotrophic lateral sclerosis”, “gastrostomy” and “prediction” in the title or abstract. Each of the studies identified applied one or two pre-specified models rather than seeking to survey a large number of models for the optimal approach. Each of the studies involved training in a single population-specific cohort and the majority did not feature external validation. Imputation of missing data was not performed in the majority of studies. No study included individualised quantitative predictions for the precise timing of gastrostomy.

Added value of this study: We provide a new model for prediction of time from diagnosis to requirement of gastrostomy for ALS patients, with best-in-class performance across a range of metrics including sensitivity, specificity and the absolute error in prediction accuracy. We provide optimised model selection, hyperparameter tuning and imputation of missing data. Importantly we have optimised imputation of missing data and hyperparameter tuning to maximise utility and to avoid overfitting, even for deep learning models. Model performance is demonstrated by training and testing in multiple cohorts including two different external validation cohorts sourced from distinct populations and clinical contexts.

Additionally we present ‘predicTTE’, which is a customizable ‘app’ and accompanying online portal for any time-to-event analysis in any disease. Our tool enables a researcher with a dataset but no bioinformatics experience to design an optimal prediction model and make it available via an online portal to end-users within a ‘data-secure’ environment. In addition we provide the capacity for end-users to directly contribute additional training data.

Implications of all the available evidence: Our tool can be used to predict the timing of future gastrostomy for ALS patients at the point of diagnosis. To enter clinical practice a prospective study will be required, but we have demonstrated stable model performance across multiple populations and clinical contexts. Our tool and online platform could be used to implement a similar strategy in other diseases.

A



B

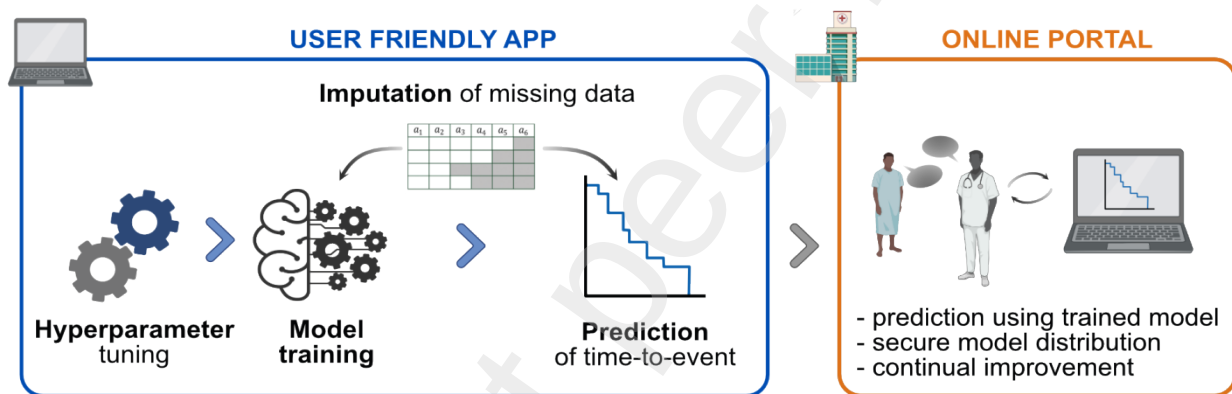


Figure 1: Accessible and optimal time-to-event prediction of gastrostomy in ALS is achieved through implementation of state-of-the-art machine learning models within an app and online platform. (A) We have developed an optimal deep-learning (DL) model for prediction of time from diagnosis to requirement of gastrostomy which achieves a clinically actionable performance that is stable across multiple external validation cohorts from different populations and clinical contexts. **(B)** To develop our model we implemented a set of deep learning and spline models for time-to-event modelling. Our pipeline includes model selection, hyperparameter tuning, model training and imputation of missing data. This functionality is provided within a fully customizable ‘app’ (left panel). An accompanying online portal includes capacity for end-users including clinicians and researchers, to access and perform predictions using a trained model, and to contribute new data for model improvement, all within a data-secure environment (right panel).

Introduction

Amyotrophic lateral sclerosis (ALS) is an incurable neurodegenerative disease where death results from motor neuron (MN) loss leading to respiratory failure. Relative to other neurodegenerative diseases ALS is rapidly progressive with the majority of patients surviving <5 years from diagnosis (1). Consequently the clinical course of ALS is largely a function of disease progression rather than other factors such as frailty and co-morbidities; this raises the possibility that, more than for other neurodegenerative diseases, ALS could be predictable based on measurements made at diagnosis.

Nutritional status and weight loss are independent prognostic factors for survival during the course of ALS (2). Dysphagia and limb weakness, combined with hypermetabolism (3) limit the utility of dietary modification to avoid weight loss. Indeed, gastrostomy is current recommended clinical practice for all ALS patients (4) who reach >5% weight loss from diagnosis (5). Gastrostomy can stabilise weight loss in ~50% of patients (5). Late gastrostomy often does not stabilise weight loss, and is also associated with higher periprocedural mortality (2). The decision to have a gastrostomy is complex but uncertainty about timing means that the decision is often made late (6,7). Timing of gastrostomy is also relevant for clinical trials: Broadly predictions regarding the rate of disease progression can be used to stratify patients and so improve statistical power by reducing heterogeneity between test and placebo groups (8). More specifically, investigational medicinal products are often given orally and therefore the need for gastrostomy is can be an exclusion criteria. Therefore there is a compelling case for the need for accurate prediction of the timing for gastrostomy requirement.

We provide a clinical tool to predict optimal timing for gastrostomy in ALS patients. We used an objective standard measure to indicate gastrostomy requirement: 5% weight loss from diagnosis (5). Cox regression (9) is a popular model for time-to-event tasks which assumes a fixed proportional-hazard ratio, whereby the relative hazard-rate between patients is invariable over time. This is an unrealistic assumption for many contexts and has likely led to misinterpretation of the underlying drivers of time-to-event. Instead, to predict time until gastrostomy requirement, we tested >5,000 different prediction model configurations including spline models (10) and non-linear deep learning (DL) models designed for time-to-event prediction (11). We show that an optimal non-linear DL model is superior in our prediction task, particularly in the accuracy of individualised predictions as opposed to purely discriminative measures such as concordance. Model training was used 3,000 patient profiles from within the PRECISION (12) dataset, taken from eight sites across Europe. Model performance is stable in internal cross-validation and in two external independent cohorts from Sweden and the United States. Model performance is robust to missing data, even when specific

measurements are entirely missing in the external validation cohorts. Our approach to prediction for timing of gastrostomy is summarised in **Fig. 1A**.

To ensure widespread usability we have implemented our model within an 'app' and accompanying online platform. The online platform includes the capacity to provide end-users such as clinicians, with secure access to the trained model for prediction, and with the option to contribute new data to improve model training. Moreover, our app is flexible and can be used to find an optimum model to make predictions for *any longitudinal event in any disease*; this functionally is automated end-to-end requiring no computational 'expertise'. We have named our software 'predicTTE' (predicting time-to-event). Our tool is summarised in **Fig. 1B**.

Results

Choice of clinical measurements for prediction of time between diagnosis and gastrostomy requirement

To train a model for prediction of the timing of future gastrostomy requirement, as indicated by 5% weight loss from diagnosis (5), we chose a set of baseline clinical variables that have previously been associated with the rate of ALS progression (10): age of disease onset, presence/absence of an ALS-associated *C9orf72* mutation (13), site of disease onset, diagnostic delay, ALSFRS-R slope, forced vital capacity (FVC, percentage of predicted based on normative values for age, sex, body height), and cohort or geographical location of clinical care. Diagnostic delay is the time from symptom onset to diagnosis with ALS and has been consistently linked to ALS survival (14), probably because it represents the speed of progression to the point where the disease is both clinically manifest and the patient has sufficient functional impairment to seek medical assistance. The ALSFRS-R is a commonly used functional rating scale for ALS (15); to infer the rate of change or 'slope' we assumed a linear decline between the time of symptom onset and the time of diagnosis i.e. over the period which constitutes the diagnostic delay. Importantly all of these data are frequently collected at ALS diagnosis. We also added sex because there is consistent evidence that sex impacts ALS biology (16), weight at diagnosis, and presymptomatic/premorbidity weight.

Cohorts used for model training and evaluation were broadly similar in the proportion requiring gastrostomy (**Supplementary Fig. 1A**), the rate of disease progression (**Supplementary Fig. 1B**), the proportion of missing data (**Supplementary Fig. 1C**) and the distribution of model covariates (**Supplementary Fig. 1D**). The exception was slightly more rapid progression in the Swedish validation cohort (**Supplementary Fig. 1B**) and the fact that presymptomatic/premorbidity weight was missing from both of the external validation cohorts (**Supplementary Fig. 1D**). Heterogeneity in cohorts is a useful test of the generalisability of

model performance. We note that there is a non-linear correlation between rate of change in weight for patients who required gastrostomy during the observed period, but no correlation at all for patients with censored data (**Supplementary Fig. 1E-F**). Therefore censored patients were not used for assessment of absolute time to gastrostomy requirement, although it is possible to use these patients to assess relative measures such as concordance.

Optimised model choice and hyperparameter tuning

Next we sought to choose an optimal model for prediction of the timing of future gastrostomy requirement using our chosen baseline clinical measurements. We implemented >5,000 different model configurations including spline models (10) and a set of non-linear DL models designed for time-to-event prediction (11). Model choice depends on the specific outcome measure; popular outcome measures used in prediction models include concordance, area under the curve of the receiver operating characteristic curve (AUROC) for a binary outcome, negative log-likelihood which is a measure of goodness-of-fit for model predictions based upon a probability distribution, and the related Brier score which applies a similar principle specifically to binary outcomes. However, while these measures may perform well in a *discrimination* between individuals with different risk of event, they do not necessarily achieve adequate *calibration*, which is a measure of how well the predicted risk of hazard matches the observed risk. Calibration is essential for a clinical prognosis where relative risk is not informative, although relative risk may be more important in the context of clinical trial stratification. Neither concordance or AUROC consider calibration. The Brier score theoretically considers both discrimination and calibration but fails to achieve good sensitivity at the cost of specificity if prevalence is low (17). We considered the median absolute error between observed and predicted time to event (MAE), but this has similar pitfalls to the Brier score; we have observed that models which optimise MAE tend to introduce a systematic bias which can compromise sensitivity. Our solution to this problem is to enable model choice and hyperparameter tuning with differential weighting of multiple outcome measures and in doing so we optimise both discrimination and calibration (**Supplementary Fig. 2, Supplementary Fig. 3**). Exhaustive hyperparameter tuning testing all possible model configurations is computationally intensive and liable to overfitting; to avoid this we implemented a Bayesian framework for model tuning (18) (**Supplementary Fig. 2**).

We applied our optimised hyperparameter tuning to our training data including 3,000 patient profiles from within the PRECISION (12) dataset, taken from eight sites across Europe. For prediction of time from diagnosis to requirement of gastrostomy a logistic hazard DL model (19) produced the optimal performance in 20% held-out test data (**Supplementary Table 1, Supplementary Fig. 2B-C**).

Prediction performance is stable in internal cross-validation and external validation cohorts

Next we evaluated performance of the optimal logistic hazard DL model for the timing of future gastrostomy requirement, via internal cross validation and then in two external cohorts which were not utilised in hyperparameter tuning. We compared the performance of the optimal logistic hazard DL model to an optimal spline model because a Royston-Parmar spline model has previously been used successfully to predict ALS survival time (10). First we trained both models and performed internal cross validation using held-out cohorts from the PRECISION dataset (n=3,000) (**Fig. 2A**, left and middle panels); before then testing in two external validation cohorts from the US (PRO-ACE, n=299) and Sweden (Karolinska, n=215) (**Fig. 2A**, right panels). For the optimal logistic hazard DL model, AUROC for the presence/absence of gastrostomy requirement at 12 months after diagnosis was 0.75 (**Fig. 2A**, upper panels and **Fig. 2D**) and concordance was 0.67 (**Fig. 2A**, middle panels) for all cohorts; and performance of the optimal spline model was not significantly different to the DL model (**Fig. 2A**, upper and middle panels). For the optimal logistic hazard DL model, in internal cross validation the MAE was 3.7 months with some variability between cohorts (**Fig. 2A**, lower panels); here the optimal logistic hazard DL model was clearly superior to the optimal spline model (**Fig. 2A**, lower panels) and this difference was statistically significant (rank sum test, $p=8e-3$). In the external validation the DL model achieved a MAE of 2.2 months in the US cohort, and 3.4 months in the Swedish cohort (**Fig. 2A**). Performance in the Swedish cohort is in-line with other cohorts despite a distinct clinical profile (**Supplementary Fig. 1B**). Performance metrics are provided in **Supplementary Table 2**.

We hypothesised that model performance might be improved by incorporating longitudinal data. We repeated model choice and hyperparameter tuning for a new model, using the same training data but with the addition of weight measurement at ~6 months (range 4-8 months based upon available data, **Supplementary Fig. 1E-F**). Again the optimal model was a logistic hazard DL model (**Supplementary Table 1**). For patients who did not require gastrostomy before six months, this longitudinal DL model significantly improved prediction performance (MAE 2.6 months, AUROC at 12 months 0.86, concordance 0.79) (**Fig. 2B,E, Supplementary Table 2**); again there was a significant difference in prediction performance for MAE between the optimal logistic hazard DL model and an optimal spline model (rank sum test, $p=0.02$). Comparison of performance of the optimal logistic hazard DL model with training and testing within individual cohorts demonstrates that MAE improves linearly as a function of the number of training samples (**Fig. 2C**) which supports efforts to add additional training data.

The distribution of time to gastrostomy is not uniform (**Fig. 2F**, upper panel). In the training cohort 32% of patients required a gastrostomy between 5 and 10 months after diagnosis and

in the external validation cohorts this proportion was 48%. If we consider both the baseline and the longitudinal model predictions together, the optimal DL model achieves MAE of 1.2 months for these patients (**Fig. 2F**, lower panel).

We conclude that an optimally trained logistic hazard DL model is able to achieve a level of performance which is accurate and stable enough to guide clinical decision making for *individual* patients with ALS.

Data missingness is frequent and can be imputed using MissForest

Reported results above include imputation of missing covariates via a random-forest model called 'MissForest' (20). Missing data is a common real-world phenomenon, which necessitates the adoption of imputation methods that can yield highly accurate results. Moreover the clinical progression profile of ALS patients with missing data is not equivalent to those without missing data (21) and thereby, a model that neglects patients with missing data does not capture the full range of ALS phenotypic variation. Numbers and proportions of missing data used in training are detailed in **Supplementary Fig. 1**.

MissForest was trained using the relationships between observed covariates in the training dataset (PRECISION, n=3,000 of whom n=626 had no missing data). In both internal cross validation in the training cohort (**Fig. 3A**, left panel), and in the external validation cohort (**Fig. 3A**, right panel), imputation of missing data improves model prediction performance as measured by MAE. Heterogeneity in missing data between cohorts prohibited evaluation using individual cohorts (**Fig. 3B**), and therefore we pooled cohorts in evaluation of model performance (**Fig. 3A,C**). We noted that presymptomatic/premorbid weight was missing almost entirely from both external validation cohorts (**Supplementary Fig. 1**). We wondered if a covariate could be entirely imputed and how this would impact model performance. Strikingly, in external validation, a model with imputed presymptomatic/premorbid weight significantly outperforms a model not trained with presymptomatic/premorbid weight, or one where missing presymptomatic/premorbid values have been replaced with the cohort mean (**Fig. 3C**). This suggests that the optimal DL model is able to learn relationships between covariates, and that imputation adds useful information for prediction performance.

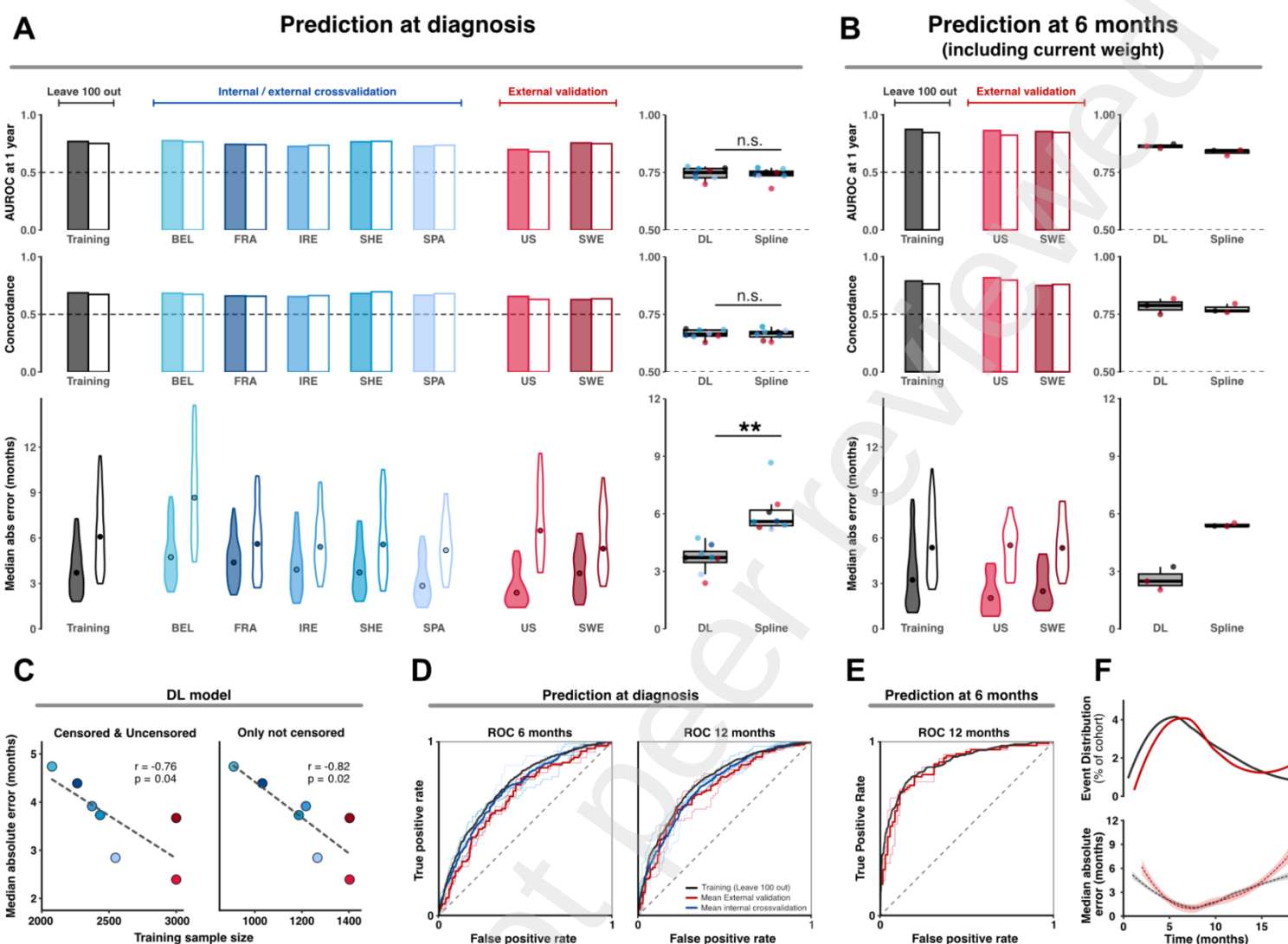


Figure 2: Prediction of time to gastrostomy is stable in internal and external validation. (A) Prediction performance for time from diagnosis to requirement for gastrostomy is evaluated in the PRECISION training dataset ($n=3,000$) via leave-100-out cross validation (left panel) and leave-one-cohort out cross validation (middle panel); and in two separate external validation cohorts (right panel). Model performance is shown for AUROC at 12 months from diagnosis (upper panels), concordance (central panels), and the median absolute error between observed and predicted time from diagnosis to requirement for gastrostomy (MAE) (lower panels). (B) For patients who did not require a gastrostomy at six months after diagnosis a new longitudinal model is trained and evaluated including observed weight at ~6 months. Violin plots are used to illustrate MAE (circle) together with the interquartile range. (C) Model performance by MAE is plotted against the number of training samples for both censored and uncensored observations. (E-F) ROC curves for the initial prediction of time from diagnosis to gastrostomy (D) and in the updated longitudinal model (E). (F) Recorded times to requirement for gastrostomy are binned by month and plotted to show the distribution through the cohort (upper panel) with the corresponding model performance by MAE (lower panel). Dark shading indicates the optimal logistic hazard deep learning (DL) model; light shading indicates the optimal spline model. BEL = Belgium, FRA = France, IRE = Ireland, SHE = UK, SPA = Spain, PRO-ACE = USA, SWE = Sweden. In all panels, colour indicates cohort.

An important decision in imputation of missing data is whether to include the outcome variable in imputation. Inclusion can increase the risk of overfitting to the training set whereas omission can artificially depress the importance of imputed data points (22). However, if the missingness of the covariate is a determinant of the relationship between the covariate and the outcome variable (e.g. if those with rapid disease progression are less likely to perform all tests) then imputing including the outcome variable relies on an incorrect assumption (23). To evaluate these alternatives we performed imputation, training and prediction under both scenarios: with and without the use of the outcome variable in the imputation of missing datapoints; here the outcome variable is the *uncensored* timing of gastrostomy requirement. In both internal cross validation in the training cohort (**Fig. 3A**, left panel) and in the external validation cohorts (**Fig. 3A**, right panel) imputation of missing data without the outcome variable improves model prediction performance.

To further test the performance of the MissForest model we randomly selected and omitted one covariate value from each of 150 patients (10% of non-censored patients) within the training dataset, in addition to existing missing data. These data were then imputed using the MissForest model and used to train the optimal DL model to predict time from diagnosis to requirement for gastrostomy. This procedure was repeated with three different random selections of covariates, but we avoided omitting two covariates from any single patient in each instance. For all covariates there was a significant correlation between predicted time to gastrostomy requirement with actual values and with imputed values (**Fig. 3D**). Moreover, imputed values are strongly correlated with the values they replaced (**Supplementary Fig. 4**) supporting the efficacy of our imputation strategy.

The full set of clinical covariates contribute to prediction model performance

We wanted to determine which covariates are important for prediction model performance. A theoretical advantage of a DL model is that it is capable of extracting knowledge from non-linear combinations of covariates. This contrasts with an additive linear model where each covariate contributes a fixed value to prediction performance. For example, applying a linear Cox proportional hazards model to prediction of time from diagnosis to requirement for gastrostomy demonstrates that performance is dictated by site of disease onset, ALSFRS-R slope and sex (**Fig. 4A**). To investigate the relationship between covariates and model performance for the optimal DL prediction model we created a simulated dataset of 150,000 patients derived from our original training data. Each of the ten covariates was permuted randomly in a replica of the original 3,000 patients, and this was repeated five times for each covariate. We could then determine the effect on model performance of altering each covariate.

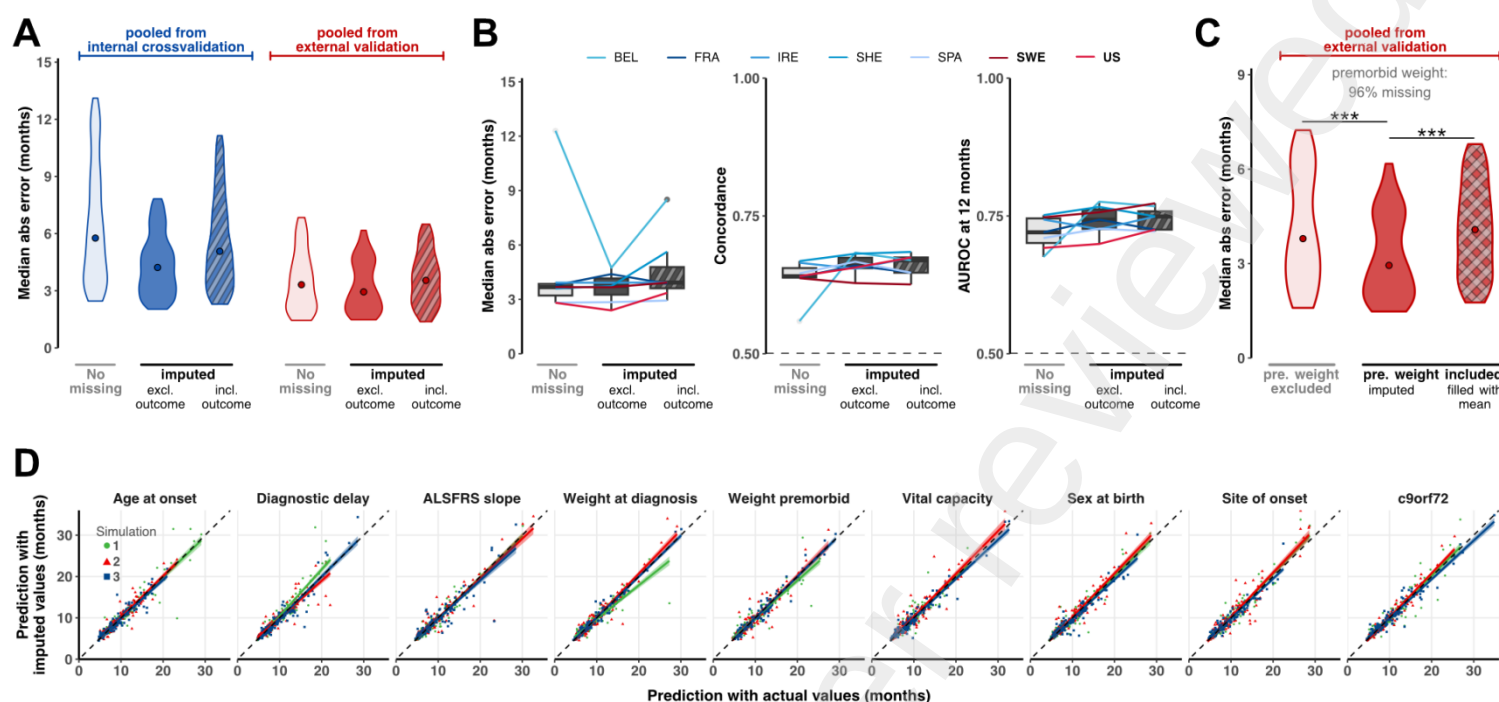


Figure 3: Missing datapoints can be imputed using MissForest without degrading prediction performance.

(A) Median absolute error between observed and predicted time from diagnosis to requirement for gastrostomy (MAE), consisting of the pooled predictions taken from leave-one-cohort-out cross validation in the training dataset (left, blue), and the pooled predictions in the external validation dataset (right, red). Performance is shown after removing all patients containing missing data (light shading) or imputing missing data (dark shading), either using no outcome measurement (no pattern) or observed event times (diagonal stripes). (B) Performance evaluated by MAE, concordance and AUROC at 12 months is shown for all 7 cohorts instead of the pooled data using the same approach as in (A). (C) In the pooled external validation dataset 96% of the patients had no record of presymptomatic/premorbid weight. The MAE is shown comparing a model, where in training and validation premorbid weight was excluded (light shading), imputed, or filled with the mean presymptomatic/premorbid weight of the training cohort. *** $P < 0.001$ rank sum test. (D) Prediction of time from diagnosis to gastrostomy requirement in the training cohort is compared using actual or imputed values of each covariate. 150 observations (10% of uncensored values in the training cohort) were randomly removed and imputed to generate predictions, and this procedure was repeated in three different simulation rounds. Predictions were performed using the optimal DL model. Predicted values are shown until 95th quantile, colors indicate the simulation round.

For all outcome measures and in both the optimum spline model and the optimal logistic hazard DL model, almost all covariates impact model performance (**Fig. 4B**). For MAE and the optimal logistic hazard DL model then *all* covariates are important (**Fig. 4B**, right upper panel).

Next, we examined the relationship between predicted time from diagnosis to gastrostomy requirement as it depends upon *combinations* of any two other covariates. The idea was to look for evidence of non-linear interactions between covariates. To do this we simulated a dataset including 187,500 data instances; simulated patients were given all possible combinations of quintiles for each of the six numerical covariates, and both possible values for binary covariates. We then generated a predicted time to gastrostomy requirement for each simulated patient and then examined how systematically altering each covariate impacted the value of the model prediction. We find evidence that both the optimal spline model and the optimal logistic hazard DL model capture interactions between covariates (**Fig. 4C**). Notably, *C9orf72* genetic status has an opposite effect on prediction of time to event in the DL model compared to the spline model (**Fig. 4D**). Using the interaction between ALSFRS-R slope or presymptomatic/premorbid weight and weight at diagnosis as an example, for the DL model there is evidence that covariate interactions are non-linear (**Fig. 4C**, upper panels) but for the spline model the interactions are linear (**Fig. 4C**, lower panels). The same is true for all combinations of covariates (**Fig. 4E,F,G, Supplementary Fig. 5, Supplementary Fig. 6**).

Webpage implementation of predicTTE

predicTTE is a self-contained software package designed to facilitate the design and training of time-to-event prediction models. We have provided access to our trained model for prediction of time to gastrostomy requirement through an accompanying online portal (www.prediccte.org) designed to be used by researchers and clinicians without any requirement for computational expertise. Users of the portal, including clinicians, can also contribute their own data for model training (**Fig. 1**) and access our app to design and train their own models. Data submitted within the portal is anonymised and securely stored.

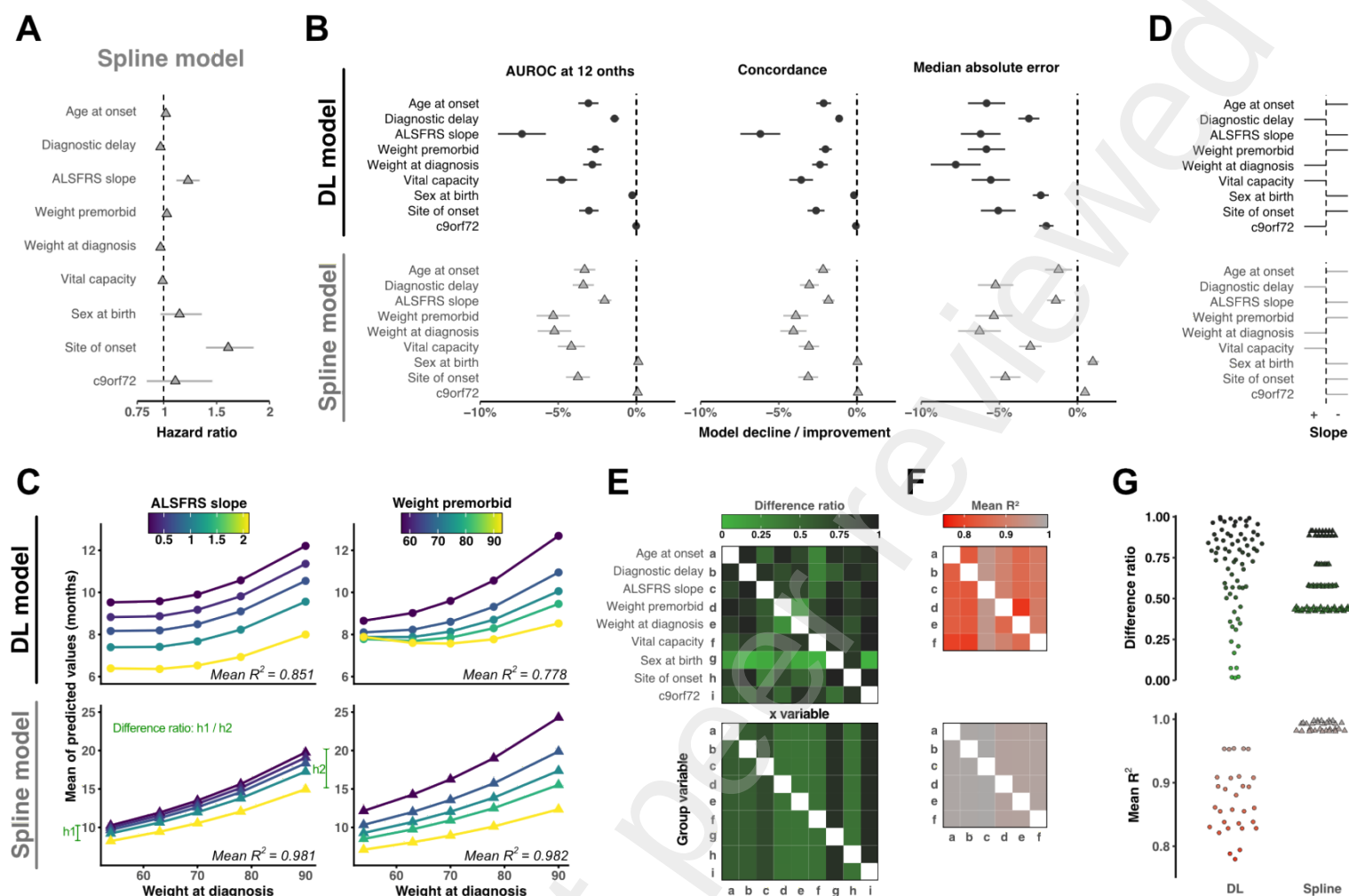


Figure 4: Optimal DL model performance is determined by a non-linear combination of covariates. (A) Hazard ratios (and 95% confidence interval) for each covariate are calculated from the optimal spline model used to predict time to requirement for gastrostomy in the training data ($n=3,000$). (B) Prediction times produced using the optimal DL model (top panels) and the optimal spline model (lower panels), were evaluated for the training dataset and 50 simulated datasets where each of the ten covariates was permuted five times. Percentage of improvement or decline in model performance was calculated by comparing to non-permuted data; positive values indicate improved model performance. Shown is the mean (circle) $\pm$ SEM (errorbar) for each of the five permutations per covariate. (C-G) A completely simulated dataset of 187,500 patients was created by combining every possible combination of observed values of the ten covariates; to achieve this numerical covariates were categorised into quintiles. The trained optimal DL and spline models were then used to make predictions for time to gastrostomy requirement using this dataset. (C) To consider the interaction between weight at diagnosis and ALSFRS-R slope (left panels) or presymptomatic/premonitory weight (right panels), mean predicted time to gastrostomy is plotted for all possible combinations of these covariates. R^2 is calculated by fitting a linear model for each combination, and the difference ratio (h_1/h_2) captures the variability in model predictions as a function of weight at diagnosis. Similar linear models are fitted for all possible covariate pairs: (D) The slope of linear models demonstrates the direction of interdependence where a negative slope indicates that a higher value of the covariate is associated with early gastrostomy requirement. (E) difference ratio and (F) mean R^2 values for all possible pairs of covariates using the optimal DL model. (G) Distribution of the difference ratios (top) and the mean R^2 -values (bottom) for both the optimal DL and optimal spline models.

Discussion

Accurate prediction to enable timely gastrostomy for ALS patients has the potential to significantly improve clinical care, optimise healthcare resource planning and improve eligibility selection in clinical trials of oral drugs. Our optimal prediction model provides individualised predictions of the timing of future gastrostomy requirement which are accurate enough to guide clinical decision making. Importantly prediction performance is stable across multiple cohorts, populations and clinical contexts suggesting that it is generalisable. Absolute prediction time could be used to provide an individualised target for gastrostomy preparation; or the binary prediction as to whether gastrostomy will be required at 12 months after diagnosis could be used to prioritise immediate versus delayed gastrostomy planning. We await a prospective study, but our model has immediate utility for clinical care and trial screening.

We chose to use 5% weight loss from diagnosis as an objective measure of the requirement for gastrostomy. Here we are following the recommendations of a prospective cohort study (5). This study noted that more than 10% weight loss from diagnosis at the time of gastrostomy was associated with reduced survival but also noted that refractory cachexia is likely to occur after 5% weight loss from diagnosis, based upon research in oncology (24). The idea of our work is to provide prediction to facilitate early intervention and therefore we focused on the 5% cut off. This will also ensure that 10% threshold is *not* reached within the error inherent in our prediction times.

We provide an implementation of a random forest model – MissForrest – for imputation of missing data. In testing this method, we considered whether the outcome variable should be used in imputation. We show that inclusion of the outcome variable can degrade model performance in a validation dataset. This is in opposition to previous findings (25) but notably the contradictory study predates the development of the set of spline and DL time-to-event prediction tools we have applied. In particular, DL models may be more vulnerable to overfitting when missing values within the training data carry an artificial signature of the outcome variable. Our missing data is sufficiently effective that imputation of presymptomatic/premorbidity weight improves model performance, even when this variable is not recorded in the external validation cohort. We also provide a new approach for hyperparameter tuning and model selection based on a weighted combination of outcome variables and Bayesian optimisation. Our method successfully avoids a compromise between discrimination and calibration.

Our optimal DL prediction model is superior to an optimal spline model in discrimination but particularly in calibration as measured by the accuracy of absolute prediction values as

measured by the MAE. We provide evidence that this is because of non-linear dependencies captured by the deep learning model which are not captured by an optimal spline model. In the context of ALS, provision of an absolute time value aligns with expressed patient preference (10). A practical advantage of an exact prediction value is that results are optimally portable to other applications such as a study of genetic drivers of ALS rate of progression.

Our predicTTE framework including our app and online portal, are designed to widen accessibility to optimal time to event prediction models. The aim is to provide cutting-edge computational tools to non-expert users who hold the appropriate clinical data. Rapid development in the statistical tools for time-to-event prediction has had limited impact because of the requirement for advanced statistical knowledge and computational expertise. Improvements in prediction will likely provide accurate prognoses, guide personalised medicine and facilitate timely clinical interventions. Moreover, our secure online portal enables sharing of data and distribution of individualised prediction to researchers and clinicians, who can then contribute their own data via the online portal. We envision a positive feedback loop leading to exponential improvements in training data size and prediction model performance.

Disclosures

Adriano Chiò serves on the editorial advisory board of Amyotrophic Lateral Sclerosis and Neurological Sciences. Adriano Chiò serves on scientific advisory boards for Mitsubishi Tanabe, Biogen, Roche, Denali Pharma, Cytokinetics, Lilly, Ferrer, Zambon Biotech, and Amylyx Pharmaceuticals, has received a research grant from Biogen and serve on Drug Safety Monitoring Board for AB Science, Corcept, and Eli Lilly. He has received research support from the Italian Ministry of Health (Ricerca Finalizzata), Regione Piemonte (Ricerca Finalizzata), Italian Ministry of University and Research (PRIN projects), University of Turin, and the European Commission (Health Seventh Framework Programme, Horizon 2020 and Horizon Europe).

Andreas Hermann has received personal fees and non-financial support from ITF Pharma, Amylyx, Biogen, NRG therapeutics and Desitin. He has received royalties from Elsevier Press and Kohlhammer.

Caroline Ingre, Neuromuscular Disease Specialist, Head of the ALS Center at Karolinska University Hospital, ALS Research Group Lead at Karolinska Institute, Founder of the National ALS registry in Sweden and the Karolinska Treatment Center, Hosts of ENCALS ALS Meeting 2024, Sweden.

Leonard van den Berg, Professor of Neurology Director, ALS Centre, University Medical Centre, Utrecht, Chair, TRICALS Consortium, The Netherlands

Mònica Povedano Panadés reports consultancies/advisory boards for Amylyx Pharmaceuticals, Biogen, Ferrer, Grifols, Italfarmaco, Mitsubishi Tanabe Pharma and Roche. Philip Van Damme reports advisory boards for Biogen, CSL Behring, Alexion Pharmaceuticals, Ferrer, QurAlis, Cytokinetics, Argenx, UCB, Muna Therapeutics, Alector, Augustine Therapeutics, VectorY, Zambon, Amylyx (paid to institution). He has received speaker fees from Biogen, Zambon and Amylyx (paid to institution). He is supported by the E. von Behring Chair for Neuromuscular and Neurodegenerative Disorders (from CSL Behring, paid to institution).

Philippe Corcia reports consultancies or advisory boards for Amylyx, Biogen, Coave Therapeutics, Cytokinetics,, Ferrer, Mitsubishi Tanabe, QurAlis, VectorY, Zambon. He is member of the Board of the Journal Amyotrophic Lateral Sclerosis and the Frontotemporal Dementias and of the Revue Neurologique.

Ammar Al-Chalabi reports consultancies or advisory boards for Amylyx, Apellis, Biogen, Brainstorm, Clene Therapeutics, Cytokinetics, GenieUs, GSK, Lilly, Mitsubishi Tanabe Pharma, Novartis, OrionPharma, Quralis, Sano, Sanofi, and Wave Pharmaceuticals.

Orla Hardiman reports consultancies/advisory boards for Biogen, Takeda, Ferrer, Novartis, Alchemab and Medici Nova. She is Editor in Chief of the Journal Amyotrophic Lateral Sclerosis and the Frontotemporal Dementias.

Juan Vazquez Costa is a member of the ERN-NMD and reports consultancies or advisory boards for Biogen and Roche.

Acknowledgments

Data used in the preparation of this article were obtained from The Pooled Resource Open-Access Clinical Research (PRO-ACE) Database. The data available in the PRO-ACE Database have been volunteered by PRO-ACE Consortium members. This work was supported by the National Institutes of Health (CEGS 5P50HG00773504, 1P50HL083800, 1R01HL101388, 1R01-HL122939, S10OD025212, P30DK116074, and UM1HG009442 to Michael P. Snyder), the Wellcome Trust (216596/Z/19/Z to Johnathan Cooper-Knock), the Swedish Research Council (2020-03485 and 2018-05973 to Eran Elhaik), the ALS Association (Ahmad Al-Khleifat and Johnathan Cooper-Knock), the Darby Rimmer Foundation (Ahmad Al-Khleifat), TargetALS (Johnathan Cooper-Knock and Eran Hornstein) and the ISCIII (PI24/01512 to Juan Vazquez Costa). Christopher J. McDermott and this work are supported by an NIHR Professor Award (NIHR301648). We also acknowledge support from the NIHR Sheffield Biomedical Research Centre (NIHR203321) and the NIHR Sheffield Clinical Research Facility. This paper was supported by the PRECISION ALS Programme, a Research

Ireland funded academic/industry research collaboration between TRICALS, Trinity College Dublin and Biogen. This research was conducted, in part, with the financial support of Research Ireland under Grant Agreement No. 20/SP/8953 and 13/RC/2106_P2 at the ADAPT Research Ireland Research Centre at Trinity College Dublin. ADAPT, the Research Ireland Research Centre for AI-Driven Digital Content Technology, is funded by Research Ireland through the Research Ireland Research Centres Programme. Data were generated from funded projects including Euromotor (259867), the JPND supported ALSCarE, SOPHIA, and BRAIN-MEND programme and MNDA AMBRoSIA. Additional support was from ALS Stichting Nederland (grant no. NMZ Biobank/PAN Studie). Harry E. McDonough, Cristopher J. McDermott, and Pamela J. Shaw are supported by the NIHR Sheffield Biomedical Research Centre (IS-BRC-1215-20017). Pamela J. Shaw is supported as an NIHR Senior Investigator (NF-SI-0617-10077). Philip Van Damme declares grants from TBM from FWO-Vlaanderen (n° T003519N), holds a senior clinical investigatorship of FWO-Vlaanderen (G077121N) and is supported by the E. von Behring Chair for Neuromuscular and Neurodegenerative Disorders, the ALS Liga België, the KU Leuven funds “Een Hart voor ALS”, “Laeversfonds voor ALS Onderzoek” and the “Valéry Perrier Race against ALS Fund”. This work was also supported by the Horizon 2020 Programme (project Brainteaser under grant agreement 101017598; project Hereditary under grant agreement 101137074), the Italian Ministry of Education, University and Research (Progetti di Ricerca di Rilevante Interesse Nazionale, PRIN 20228N7573). This study was performed under the Department of Excellence grant of the Italian Ministry of University and Research to the “Rita Levi Montalcini” Department of Neuroscience, University of Torino, Italy. This study represents independent research part funded by the National Institute for Health Research (NIHR) Biomedical Research Centre at South London and Maudsley NHS Foundation Trust and King’s College London. Ammar Al-Chalabi is an NIHR Senior Investigator (NIHR202421). The Joint Programme on Neurodegenerative Disease (JPND) have funded data collection for patient registries over several decades along with the Charity Research Motor Neuron (RMN), the Irish Health Research Board (HRB) in Ireland, the Ulla-Carin Lindqvist Foundation in Sweden, and Fundación Miquel Valls in Spain, which played a crucial role in collating the dataset used in this paper. The MND Register of England, Wales and Northern Ireland is funded by the MND Association, with additional support through an EU Joint Programme - Neurodegenerative Disease Research (JPND) project under the aegis of JPND - www.jpnd.eu (United Kingdom, Medical Research Council (MR/L501529/1; MR/R024804/1) and Economic and Social Research Council (ES/L008238/1)) and through the My Name’s Doddie Foundation, and Alan Davidson Foundation. This paper was made possible through the collaboration of the TRICALS Consortium through the work of the PRECISION ALS Programme. Research Ireland (RI) and Biogen provided financial contributions, which supported the data processing and

analysis phases of this study. Biogen were given a chance to review and provide comments to this paper prior to submission. We would like to thank the people with MND who provided their data for this study by consenting to their inclusion. Figures were created using BioRender.com.

Methods

Study Cohorts

PRECISION: The ALS patients included in this study as part of the PRECISION cohort were recruited at specialised neuromuscular centres in the UK, Belgium, Germany, Ireland, Italy, Spain, Turkey, and the Netherlands (12). We removed all patients who received gastrostomy within 30 days of diagnosis on the basis that they were likely to meet criteria for requirement of gastrostomy at baseline and therefore prediction was not necessary. We also removed patients where the first measured weight was more than three months distant from the date of diagnosis. This left $n=3,000$ patients which were used for model training. An additional PRECISION cohort recruited in Sweden became available through the course of this project and was added as an external validation cohort ($n=215$).

PRO-ACE: The ALS patients included in this study as part of the PRO-ACE external validation cohort were recruited at sites in the United States (US) (26). Removal of patients who required gastrostomy at baseline left $n=299$ patients.

Longitudinal measurements of weight were used to calculate time to requirement for gastrostomy as indicated by 5% weight loss from diagnosis (5). To avoid confounding by the effect of gastrostomy, all weight measurements performed after the placement of gastrostomy as indicated by the ALSFRS-R score were removed. Moreover, to minimise confounding by inaccurate measurements of weight we determined the distribution of weight change over time and removed all measurements above the 95-percentile indicating an implausible rate of change in weight. A similar process was used to remove implausible outliers for all covariates.

Hyperparameter tuning and model choice

Model choice and hyperparameter tuning were guided by comparative testing of model configurations encompassing the full range of parameters within the pycox (11) implementation of MTLR (27), PC-Hazard (28), PMF (28), logistic hazard (19), DeepSurv(29), CoxTime (11), CoxCC (11), and DeepHit (30) deep learning models. Model scheme and layer structure were prioritised based upon an initial grid search followed by a Bayesian optimisation of all other parameters (18) (**Supplementary Fig. 2**). A Bayesian approach reduces the number of iterations and the potential for overfitting, particularly the use of an including an

AdamWR optimizer with decoupled weight decay in the final stage (**Supplementary Fig. 2**). Hyperparameter tuning was performed via nested cross-validation: We separated the data into 80% for training and 20%, which we designate the testing dataset, for the final assessment of model performance. We then performed 10-fold cross validation within the 80% of data designated for training; here the data was *further* divided, on 10 separate occasions, into 80% for training and 20% for validation. The final prediction is output as the median of the 10 different predictions and compared with the testing dataset. We selected a random starting seed for each of the 10 rounds of cross-validation.

Similarly we tested 20 flexible parametric models including an implemented version of the Royston-Parmar spline model, by extending the CoxPHFitter python package that allows prediction using the hazard scale with the odds scale analogous to the flexsurv R package (31). Comparing model performances on both scales using the extended python and R package resulted in a strong correlation $r > 0.99$. The optimum spline model, which was implemented for comparison of prediction performance with the optimum DL model, was using the odds scale with two nodes, analogous to (11).

In total hyperparameter tuning and model choice involved testing 632 model configurations per model type (**Supplementary Fig. 1**). Model evaluation is performed by combining two outcome measurements: the median absolute error (MAE) between actual and observed time to event, and the AUROC at 12 months after diagnosis. Models are ranked by performance in each outcome measure, and the lowest rank across all three measures is used to direct model choice. At all stages we 3-fold up-weighted the MAE because otherwise the model down-weights this due to quantitatively larger differences. Compared to the use of two outcome measures, hyperparameter tuning focused on concordance *only* does not significantly improve concordance or AUROC, but does degrade MAE performance (**Supplementary Fig. 3**). However, the predicTTE app provides capacity to select alternative outcome measures and/or weighting in future use-cases.

Training and testing optimal models

After model choice and hyperparameter tuning, the optimal model (determined by hyperparameter tuning) was trained using 10-fold cross validation similar to in model selection and hyperparameter tuning. In each fold 80% of training samples were selected at random for model training and 20% of training samples are used for evaluation. After 10 folds, the final prediction is output as the median of the 10 different predictions. The C-index (concordance) was calculated based on (32) and in case of ties in predictions and event times adjusted

according to (11). Model evaluation was ultimately carried out in the two external validation cohorts.

The baseline model for prediction of time from diagnosis to requirement for gastrostomy was trained using age of disease onset, presence/absence of an ALS-associated *C9orf72* mutation (13), weight at diagnosis and presymptomatic/premorbidity weight, site of disease onset, diagnostic delay, ALSFRS-R slope, forced vital capacity (FVC, percentage of predicted based on normative values for age, sex, body height), cohort or geographical location of clinical care, and sex. For the longitudinal model which was retrained for patients who did not require a gastrostomy at 6 months after diagnosis, additional covariates included were the slope of the change in the weight and the timing of a weight measurement at approximately 6 months (range 4 and 8 months) after diagnosis.

Training and testing the MissForest model for imputation of missing data

A key aspect of our platform is the capability to impute missing data using a model called “MissForest”, which has shown superior performance in real-world testing (33) and an ability to simultaneously handle continuous and categorical data (20). MissForest imputes data iteratively, starting with the variable with the least missing observations and progressing to the variable with the most missing observations. A random forest model is fit on the observed values. Each imputed value in this study relied on the mean result from 10 rounds of imputation because this represented the best compromise between computational time and independence of initial random seeds. For testing predictions within the external validation dataset, to avoid data leakage missing covariates for each patient were imputed using a new dataset including the training dataset and *only* the specific single patient from the validation dataset. As an extra test we randomly selected and omitted 150 data points from each covariate in the training dataset although we avoided removing more than two data points from any single patient; to avoid bias from incorrect information this analysis only included uncensored patients (**Supplementary Fig. 4**).

Software availability

The predicTTE online portal (<https://www.predictte.org/>) includes instructional material and links to download the app for different platforms.

Supplementary Material Legends

Supplementary Figure 1: Training and external validation cohorts used for training and testing model prediction. (A) Overview of patient numbers across all cohorts. The training cohort (black) comprises seven sub-cohorts, with the upper five used for internal cross-validation (blue). Two external validation cohorts are shown in red. Dark shading indicates the number of uncensored patients where time to gastrostomy requirement was *not* observed, while light shading represents censored cases. Overall, approximately half of all patients were censored. (B) Kaplan–Meier curves for each cohort, color-coded as in (A). The right panel focuses on the critical time window during the first two years post-diagnosis. The two external validation cohorts span both extremes of survival, suggesting good generalizability. (C) Overview of missing data across all covariates and cohorts. Dark shading indicates missing values. Vital capacity and presymptomatic/premorbidity weight show notably higher rates of missingness, with premorbidity weight missing in ~96% of patients in both external validation cohorts. (D) Distributions of covariates across cohorts. Dashed lines represent the full data range, error bars indicate the interquartile range, black circles mark the median, and red diamonds denote the mean. “Site of onset” was treated as a categorical variable, while “sex at birth” and *C9orf72* mutation status were considered binary. Category proportions are color-coded as shown in the legend. (E, F) Left panels: Correlation between time to gastrostomy (x-axis) and weight slope—defined as the difference in weight from diagnosis to the closest available measurement at 6 months—for the training cohort (top, blue) and the external validation cohort (bottom, red). Uncensored patients are shown in blue/red, and censored patients in gray. Only patients who underwent gastrostomy or who passed the 6-month mark were included in the updated model (refer to **Fig. 2B**). Right panels: Timing of the closest weight assessment to the 6-month mark, color-coded by the weight slope. On average, weight measurements were taken around the 6-month point, with 50% of patients assessed between 5 and 7 months and 95% of all patients assessed between 4 and 8 months. There was no clustering of slope values based on assessment timing, supporting the validity of using the nearest measurement to 6 months in the longitudinal model.

Supplementary Figure 2: Hyperparameter tuning and model choice (A) Model selection flowchart. The tuning consists of three steps. First grid tuning where only 3 variables are varied but they are varied exhaustively, namely layer structure, the scheme and the intervals. The best three combinations of these variables are selected for the second step where 6 variables are optimised using a Bayesian approach: Optuna (18). Here 100 combinations are tested using an Adam optimizer. In the third step, 3 more variables are optimised (9 total) and an AdamWR optimiser is employed to test 500 combinations. Optuna aims to minimise a

combined feature consisting of the MAE and the AUROC at 12 months where the MAE is 3x upweighted compared to the AUROC. All three steps are repeated for each model type and finally, the best model configuration is selected for training and prediction. **(B)** Comparison of performance in different outcome measures for all DL model types demonstrating that the optimal logistic hazard DL model performs best in a combination of all three measures. Density plots shown top and right. **(C)** Model performance over the three steps of hyperparameter tuning. Left panel: Choice of covariate combinations selectively improves performance over time (left to right) demonstrating that Optuna iteratively improves model performance without exhaustively testing all possible combinations of variables. Colour indicates DL model type as in **(B)**. Right panel: Boxplot showing the same data grouped by the three steps of tuning, indicating the positive tendency and improvement from one step to the other, mainly for the chosen logistic hazard DL model and in particular with respect to MAE.

Supplementary Figure 3: Impact of outcome-based model selection on predictive performance across validation cohorts. Using a fixed set of model configurations (see **Methods; Supplementary Figure 2**), the final model was selected based either on a compromise across multiple outcome metrics—median absolute error (MAE), concordance index (C-index), and AUROC at 12 months—via the TOPSIS algorithm (dark shading), or based solely on maximising the concordance (light shading). Both resulting model configurations were evaluated using leave-one-cohort-out and external validation. Line colours represent individual cohorts, consistent with **Fig. 2**. While selecting the model based on concordance marginally improved both concordance and AUROC at 12 months, it led to a significant increase in MAE. These findings underscore that the choice of outcome metric used for model selection meaningfully influences final predictive performance, and that model selection strategies should be guided by specific clinical or research priorities.

Supplementary Figure 4: Effect of pairwise covariate interactions on predicted time to gastrostomy requirement. Each panel displays the mean predicted time to gastrostomy requirement (y-axis) from a simulated cohort of 187,500 patients, stratified across pairwise combinations of six numerical and three categorical covariates. For each panel, the x-axis represents one covariate, and the color indicates another, with points connected by lines to demonstrate the degree of linearity. Numerical covariates were represented using their empirical deciles, and categorical covariates by all observed levels. Colors for numerical covariates follow a continuous Viridis scale, while categorical variables use a fixed color scheme. Predicted times to gastrostomy requirement were generated using either the optimal DLmodel **(A)** or the optimal spline model **(B)**.

Supplementary Figure 5: Agreement between original and imputed numerical values.

Scatter plots illustrate the relationship between original (x-axis) and imputed (y-axis) values for all six numerical covariates, across three independent simulation rounds as in **Fig. 3**. Only observations with artificially introduced missing values (in test sets) and known ground truth (from the original dataset) are shown. Each point corresponds to a single observation, colored by test repetition. Solid lines represent least-squares regression fits; dashed lines denote the identity line ($y = x$) for reference. Across all features a significantly positive correlation was observed in each repetition (overall mean correlation coefficient $r = 0.543$, ranging from $r = 0.33$ (age) to 0.79 (presymptomatic/premorbidity weight)).

Supplementary Table 1: Hyperparameters for optimal prediction models

Supplementary Table 2: Performance metrics for all trained prediction models

References

1. Cooper-Knock J, Jenkins T, Shaw PJ. Clinical and Molecular Aspects of Motor Neuron Disease [Internet]. Vol. 2, Colloquium Series on Genomic and Molecular Medicine. 2013. p. 1–60. Available from: <http://dx.doi.org/10.4199/c00093ed1v01y201309gmm004>
2. Castanheira A, Swash M, De Carvalho M. Percutaneous gastrostomy in amyotrophic lateral sclerosis: a review. *Amyotroph Lateral Scler Frontotemporal Degener*. 2022 May;23(3-4):176–89.
3. Dupuis L, Pradat PF, Ludolph AC, Loeffler JP. Energy metabolism in amyotrophic lateral sclerosis. *Lancet Neurol*. 2011 Jan;10(1):75–82.
4. EFNS Task Force on Diagnosis and Management of Amyotrophic Lateral Sclerosis, et al. EFNS guidelines on the clinical management of amyotrophic lateral sclerosis (MALS)-revised report of an EFNS task force.
5. Gastrostomy in patients with amyotrophic lateral sclerosis (ProGas): a prospective cohort study. *Lancet Neurol*. 2015 Jul;14(7):702–9.
6. White S, O’Cathain A, Halliday V, Bradburn M, McDermott CJ. Supporting people with Motor Neuron Disease (MND) to make decisions about gastrostomy feeding tube placement: a survey of UK healthcare professionals’ practice and beliefs. *Amyotroph Lateral Scler Frontotemporal Degener*. 2024 May;25(3-4):290–8.
7. White S, O’Cathain A, Halliday V, Croot L, McDermott CJ. Factors influencing decisions people with motor neuron disease make about gastrostomy placement and ventilation: A qualitative evidence synthesis. *Health Expect*. 2023 Aug;26(4):1418–35.
8. Berry JD, Taylor AA, Beaulieu D, Meng L, Bian A, Andrews J, et al. Improved stratification of ALS clinical trials using predicted survival. *Ann Clin Transl Neurol*. 2018 Apr;5(4):474–85.
9. Cox DR. Regression models and life-tables. *J R Stat Soc*. 1972 Jan;34(2):187–202.
10. Westeneng HJ, Debray TPA, Visser AE, van Eijk RPA, Rooney JPK, Calvo A, et al. Prognosis for patients with amyotrophic lateral sclerosis: development and validation of a personalised prediction model. *Lancet Neurol*. 2018 May 1;17(5):423–33.
11. Kvamme H, Borgan Ø, Scheel I. Time-to-Event Prediction with Neural Networks and Cox Regression [Internet]. *arXiv [stat.ML]*. 2019. Available from: <http://arxiv.org/abs/1907.00825>
12. McFarlane R, Galvin M, Heverin M, Mac Domhnaill É, Murray D, Meldrum D, et al. PRECISION ALS-an integrated pan European patient data platform for ALS. *Amyotroph Lateral Scler Frontotemporal Degener*. 2023 Aug;24(5-6):389–93.
13. Cooper-Knock J, Hewitt C, Highley JR, Brockington A, Milano A, Man S, et al. Clinico-pathological features in amyotrophic lateral sclerosis with expansions in C9ORF72. *Brain*. 2012 Mar;135(Pt 3):751–64.
14. Knibb JA, Keren N, Kulka A, Leigh PN, Martin S, Shaw CE, et al. A clinical tool for predicting survival in ALS. *J Neurol Neurosurg Psychiatry*. 2016 Dec;87(12):1361–7.
15. Cedarbaum JM, Stambler N, Malta E, Fuller C, Hilt D, Thurmond B, et al. The ALSFRS-R: a revised ALS functional rating scale that incorporates assessments of respiratory

- function. BDNF ALS Study Group (Phase III). *J Neurol Sci.* 1999 Oct 31;169(1-2):13–21.
16. Manjaly ZR, Scott KM, Abhinav K, Wijesekera L, Ganesalingam J, Goldstein LH, et al. The sex ratio in amyotrophic lateral sclerosis: A population based study. *Amyotroph Lateral Scler.* 2010 Oct;11(5):439–42.
 17. Assel M, Sjoberg DD, Vickers AJ. The Brier score does not evaluate the clinical utility of diagnostic tests or prediction models. *Diagn Progn Res.* 2017 Dec 2;1(1):19.
 18. Akiba T, Sano S, Yanase T, Ohta T, Koyama M. Optuna. In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* [Internet]. New York, NY, USA: ACM; 2019. Available from: <http://dx.doi.org/10.1145/3292500.3330701>
 19. Gensheimer MF, Narasimhan B. A scalable discrete-time survival model for neural networks. *PeerJ.* 2019 Jan 25;7:e6257.
 20. Stekhoven DJ, Bühlmann P. MissForest—non-parametric missing value imputation for mixed-type data. *Bioinformatics.* 2011 Oct 28;28(1):112–8.
 21. White IR, Carlin JB. Bias and efficiency of multiple imputation compared with complete-case analysis for missing covariate values. *Stat Med.* 2010 Dec 10;29(28):2920–31.
 22. Landerman LR, Land KC, Pieper CF. An Empirical Evaluation of the Predictive Mean Matching Method for Imputing Missing Values. *Sociol Methods Res.* 1997 Aug 1;26(1):3–33.
 23. Allison PD. Multiple Imputation for Missing Data: A Cautionary Tale. *Sociol Methods Res.* 2000 Feb 1;28(3):301–9.
 24. Fearon K, Strasser F, Anker SD, Bosaeus I, Bruera E, Fainsinger RL, et al. Definition and classification of cancer cachexia: an international consensus. *Lancet Oncol.* 2011 May;12(5):489–95.
 25. von Hippel PT. 4. Regression with Missing Ys: An Improved Strategy for Analyzing Multiply Imputed Data. *Sociol Methodol.* 2007 Aug 1;37(1):83–117.
 26. Sherman A, Berry J, Gotkine M, Glass J, Lunetta C, Floeter MK, et al. PRO-ACE: Global collaborative ecosystem in a post-PRO-ACT era (P4.6-006). *Neurology* [Internet]. 2019 Apr 9;92(15_supplement). Available from: http://dx.doi.org/10.1212/wnl.92.15_supplement.p4.6-006
 27. Fotso S. Deep Neural Networks for Survival Analysis Based on a Multi-Task Framework [Internet]. *arXiv [stat.ML]*. 2018. Available from: <http://arxiv.org/abs/1801.05512>
 28. Kvamme H, Borgan Ø. Continuous and discrete-time survival prediction with neural networks [Internet]. Vol. 27, *Lifetime Data Analysis*. 2021. p. 710–36. Available from: <http://dx.doi.org/10.1007/s10985-021-09532-6>
 29. Katzman JL, Shaham U, Cloninger A, Bates J, Jiang T, Kluger Y. DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Med Res Methodol* [Internet]. 2018 Dec;18(1). Available from: <http://dx.doi.org/10.1186/s12874-018-0482-1>
 30. Lee C, Zame WR, Yoon J, van der Schaar M. DeepHit: A Deep Learning Approach to Survival Analysis With Competing Risks. In: *Thirty-Second AAAI Conference on Artificial*

Intelligence [Internet]. 2018 [cited 2021 Nov 12]. Available from:
<https://www.aaai.org/ocs/index.php/AAAI/AAAI18/paper/viewPaper/16160>

31. Jackson CH. flexsurv: A Platform for Parametric Survival Modeling in R. J Stat Softw [Internet]. 2016 May 12;70. Available from: <http://dx.doi.org/10.18637/jss.v070.i08>
32. Antolini L, Boracchi P, Biganzoli E. A time-dependent discrimination index for survival data. Stat Med. 2005 Dec 30;24(24):3927–44.
33. Dighe AS. Clinical Decision Support: Tools, Strategies, and Emerging Technologies, An Issue of the Clinics in Laboratory Medicine. Elsevier Health Sciences; 2019. 153 p.